2) LENGTH OF A CURVE

* The length of the arc of the curve $y=f(x)$ between $x=a$ and $x=b$ is given by:

$$S=\int_a^b \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$$

← If the curve is cartesian ie $y=f(x)$

$\frac{dy}{dx}=f'(x)$

** If the curve is parametric $\begin{cases} y=g(t) \\ x=f(t) \end{cases}$ between $t=t_1$ & $t=t_2$ length of curve is:

$$S=\int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2}\,dt$$

← If the curve is parametric:

$x=g(t) \rightarrow \frac{dx}{dt}=g'(t)$

$y=f(t) \rightarrow \frac{dy}{dt}=f'(t)$

Example 1

Find the length of the curve $y^2=x^3$ between $x=0$ and $x=4$.

Solution:

$y^2=x^3$ ie $y=x^{3/2}$ $\therefore \frac{dy}{dx}=\frac{3}{2}x^{1/2}$, $1+\left(\frac{dy}{dx}\right)^2=1+\frac{9}{4}x$

So that:

$$S=\int_0^4 \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx=\int_0^4 \sqrt{1+\frac{9}{4}x}\,dx$$

$$=\int_0^4 \left(1+\frac{9}{4}x\right)^{1/2}dx=\frac{2}{3}\cdot\frac{4}{9}\left(1+\frac{9x}{4}\right)^{3/2}\Bigg|_{x=0}^{x=4}$$

$$=\frac{8}{27}\left[10^{3/2}-1\right]$$

$$\therefore S \approx 9.07 \text{ units}$$

Example 2

Find the length of the curve $x = 5(2t - \sin 2t)$, $y = 10\sin^2 t$ between $t = 0$ and $t = \pi$.

Solution:

$x = 5(2t - \sin 2t)$, then $\dfrac{dx}{dt} = 5(2 - 2\cos 2t) = 10 - 10\cos 2t$

$y = 10\sin^2 t$, then $\dfrac{dy}{dt} = 20\sin t\cos t = 10\sin 2t$

we have:

$$\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 100 + 100\cos^2 2t - 200\cos t + 100\sin^2 2t$$

$$= 100 + 100(\cos^2 2t + \sin^2 2t) - 200\cos 2t$$

$$= 200 - 200\cos 2t$$

$$= 200(1 - \cos 2t)$$

$$= 400\sin^2 t, \quad \because \cos 2t = 1 - 2\sin^2 t$$

$$\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = 20\sin t$$

Length of arc:

$$S = \int_0^\pi \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

$$= \int_0^\pi 20\sin t\, dt$$

$$= 20\left[-\cos t\right]\Big|_0^\pi$$

$$= 20\left[-(-1) - (-1)\right]$$

$$\therefore S = 40 \text{ units}$$